

What to Expect During the Root KSK Rollover

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Executive Summary

In October 2026, a rollover of the Root Zone KSK will take place that provides an important security update to the Internet's Domain Name System (DNS). A very small percentage of Internet users may encounter problems in resolving some domain names as a result of this change. This document describes which users may encounter problems and details the kind of issues they will see at various times.

In general:

- Users who rely on software or network operators that do not check the integrity of DNS responses (known as "DNSSEC validation") will not see any effect.
- Users who rely on software or network operators that use the new KSK configuration will not see any effect from the rollover.
- Users will likely start seeing the effects at some point in the 48 hours after the rollover happens if their software or network operator's configurations have not been properly updated.
- It is impossible to predict when affected operators will notice that DNS lookups are failing for them.
- The vast majority of users are expected to be unaffected.

1. Introduction

The cryptographic key that is used to secure the DNS root zone (the "Root Zone KSK") is planned to change in October 2026. This change process, known as a rollover, has been successfully conducted once before in October 2018. This second planned rollover was first announced in 2024[1].

Following the rollover in 2018, ICANN and the community undertook work to establish a regular rollover cadence, a process delayed by the COVID-19 pandemic, resulting in the "Proposal for Future Root Zone KSK Rollovers",[2] which set a standard three-year cycle. This rollover is the first to follow that schedule. This document is part of ICANN's ongoing effort to prepare the global Internet community for the transition, focusing on what is expected in the period immediately following the rollover.

This document has three primary audiences:

- Operators of software that validates the security of DNS responses (known as "validating resolvers") who want to know what to look for once the rollover happens
- Nontechnical press and others who intend to write about the rollover before and as it happens
- Researchers who will be monitoring the DNS for indications of resolver failure after the rollover happens

[1] https://www.iana.org/news/2024/update-dnssec-ta
[2] https://www.icann.org/en/announcements/details/proposal-for-future-root-zone-ksk-rollovers-1-11-2019-en

This document is of limited interest to users whose resolvers are already prepared for the rollover or who do not perform DNSSEC validation; for them, the rollover will pass unnoticed.

The rollover is scheduled for 11 October 2026 and seeks to replace the cryptographic key with a new one codenamed “KSK-2024”.

KSK-2024 was first pre-published into the root zone in January 2025, providing approximately 21 months of advanced notice before activation. This extended pre-publication period provides substantial time for resolvers to update their trust anchors automatically using processes described in RFC 5011. It is a longer pre-publication period than for the first rollover in 2018.

Sections 2 and 3 of this document describe what will happen after the rollover to validating resolvers that are prepared for the rollover, and to those that are not. Section 4 describes what might be seen by researchers who are monitoring traffic to the DNS Root Server System. Throughout this document, language describes outcomes that may vary by resolver configuration. Only a resolver's operator can know with certainty what software is running and how it is configured.

1.1 Definition of the Root KSK Rollover

The DNS root zone was first signed with DNSSEC in 2010. The DNS root zone has two types of keys: zone-signing keys (ZSKs) that sign the main data in the root zone and key-signing keys (KSKs) that sign just the root key set (both ZSKs and KSKs) in the root zone. Every three months, a new ZSK is published. Each new ZSK is signed by a longer-lived KSK.

The rollover occurs when the root KSK is changed and the new KSK starts signing the root key set for the zone. The key being retired is codenamed KSK-2017, which has signed the root key set since the first rollover in October 2018. After the rollover, KSK-2017 will no longer be signing the root key set: Instead, KSK-2024 will be signing the root key set.

1.2 Trust Anchors

In order to understand how the rollover will happen, it is also important to understand how the validating resolver trusts the root KSK. Each validating resolver is configured with a set of trust anchors, which are copies of the keys or key identifiers that match the Root Zone KSK. Trust anchors are typically configured automatically by software vendors or by the resolvers which are configured to automatically update the trust anchors using the process described in RFC 5011[3] or by the resolver operator who manually adds a new KSK to the resolver's trust anchor store.

Before KSK-2024 existed, all validating resolvers had KSK-2017 configured as a trust anchor. After KSK-2024 was generated, resolver operators need to have either manually added KSK-2024 to their resolver's trust anchor configuration, used an automated

[3] https://datatracker.ietf.org/doc/rfc5011/

DNS-based update mechanism to update their configuration (RFC 5011), or updated their configuration using software updates from their software vendor. The new key will have been pre-published for 21 months, providing ample time for this propagation to occur. Resolver operators who have not yet added KSK-2024 to their trust anchor configuration will be unable to validate responses after the rollover.

Important note to resolver operators: All operators of validating resolvers reading this document should immediately verify whether they are prepared for the rollover by checking their current trust anchors. If operators are not ready, they should update to the latest trust anchors at their earliest opportunity. Operators of resolvers that are not performing DNSSEC validation need take no action as they are unaffected by the rollover.

Note on the future algorithm rollover: This rollover replaces KSK-2017 with KSK-2024, both of which use the same cryptographic algorithm (RSA/SHA-256) and key size (2048 bits). It is expected the next rollover, estimated to conclude in 2029, will also change the cryptographic algorithm (to ECDSA). This is a distinct event from the rollover happening in 2026 and is not covered by this document.[4]

2. Resolvers That Are Prepared for the Rollover

Resolvers that are prepared for the rollover already have KSK-2024 configured as a trust anchor. When the rollover happens, these resolvers will continue to work just as they did before the rollover because the new root KSK is already trusted to sign the root key set. Some resolver software might note in the operational logs that a rollover happened, but those log entries are unlikely to be seen unless the operator is specifically looking for them.

Users of resolvers that are prepared for the rollover will see no difference when the rollover happens. The responses they get to normal queries will be identical before and after the rollover. Early adoption data shows trust anchor update rates tracking closely with those observed during the first rollover,[5] and the vast majority of users whose resolvers perform DNSSEC validation are expected to be using resolvers that are prepared for the rollover.

Many Internet users have more than one DNS resolver configured. If any of the resolvers that a user has configured is prepared for the rollover, the user's software should find that resolver after the rollover and continue to use it. This might slow down DNS resolution as their system keeps trying the resolver that is not prepared before switching to the resolver that is prepared, but the user will still get DNS resolution.

[4] https://www.icann.org/resources/pages/ksk-algorithm-rollover-en
[5] https://blog.verisign.com/security/2024-2026-root-zone-ksk-rollover-initial-observations/

3. Resolvers That Are Unprepared for the Rollover

A resolver with only KSK-2017 configured as a trust anchor will begin to fail validation after the rollover, but when that failure first occurs is not predictable. Although the key change in the DNS is instantaneous, there is a time lag before each resolver sees it. Each record in the DNS has a "time-to-live" (TTL) value which specifies how long resolvers should cache the response without rechecking if it has changed. During this time a resolver will not try to get a newer version of the record. After the moment of the rollover, a resolver will likely still have a cached copy of the root DNSKEY records, validated while KSK-2017 was still active, and will therefore continue to validate successfully, at least for a while.

3.1 Failure Starts When the ZSK Cannot Be Validated

Every time a validating resolver gets a response from an authoritative name server, it checks the signature on that response. It saves the validation status of the signature on each name in its cache. To validate the signature on a name like "www.example.com", the resolver needs to validate the signatures on the root, on ".com", on "example.com", and "www.example.com". Resolvers typically cache these validations, so they don't perform them for each name. Most resolvers only perform validations when the validation status might have changed.

The TTL for the KSK and ZSK records is 48 hours. If a resolver obtains the root key set and validates it just before the rollover happens, that resolver may not know about the rollover for almost two days, because the resolver will not fetch a new KSK until it gets the first query after the TTL of root key set has expired. In a normal resolver with only a few users, that triggering query will happen within a few minutes (or even seconds) after the DNSKEY records' TTL has expired. On a resolver with a single user, the time before the first query could be hours, or even days, after the root key set's TTL has expired.

Note that this description is a bit simpler than what really happens. For example, some resolvers enforce a maximum length on TTLs, which could make those resolvers see the key rollover in a shorter period of time. Other configuration choices can also affect when the resolver first sees the rollover.

3.2 What Users Will See When All of Their Resolvers Fail

At some point within 48 hours after the rollover, some users' DNS queries will begin to fail when the resolver is triggered to fetch the root key set again. As explained above, one cannot predict when the first results will fail during that 48-hour period.

When this failure happens, if the user has multiple resolvers configured, their system software may try the other resolvers that the user has configured. This might slow down DNS resolution as their system keeps trying the resolver that is not prepared before switching to the resolver that is prepared, but the user will still get DNS resolution and might

not even notice the slowdown. However, if all the user's resolvers are not prepared for the rollover (such as if they are all managed by one organization and that organization has not made any of their resolvers ready), the user will start seeing failure sometime in the 48 hours after the rollover.

Users will see different symptoms of failure depending on what program they are running and how that program reacts to failed DNS lookups. In browsers, it is likely that a web page will become unavailable or appear only partially complete (e.g., images on an already displayed web page might fail to appear). In email programs, users might not be able to receive new mail, or parts of message bodies may show errors. The failures will cascade until no program is able to show new information from the Internet.

Note that the term "users" here does not just indicate humans. Automated systems using unprepared resolvers for DNS resolution will also start to fail.

Many applications, including containerized services and software using DNS-over-HTTPS or other private stub resolver configurations, may use resolvers configured at build or deployment time, independently of the host operating system's resolver settings. In these cases, the resolver in use may not be an explicit operational choice and diagnosing failures will require identifying which resolver the application has been configured to use.

After the operator of the resolver fixes the inability to validate, the users' Internet experience should go back to normal almost immediately.

3.3 How Resolver Operators Will Know About the Failure

An operator of a resolver who has configured their system monitoring software to look for significant errors will be alerted immediately after the resolver fetches a new copy of the root key set and it is unable to be validated. Such monitoring offers the operator the best chance of quickly detecting and recovering from the failure.

If the operator is not actively monitoring for significant errors, they will probably not know about the failing validation until automated systems that rely on the resolver start failing, or users start calling them about outages. If the operator is also only using resolvers with incorrect trust anchor configurations, they may be unable to receive email and might only hear about the problems by phone.

Failures will not be uniform across all users. Two users attempting to reach the same host at the same time may get different results if they are using different resolvers; one may succeed while the other fails. Where a user's system or network routes queries to a mix of prepared and unprepared resolvers, failures may also be intermittent for the same user. Either way, this inconsistency can make the root cause harder to identify.

3.4 Recovering from Being Unprepared

As soon as operators discover that their resolver's DNSSEC validation is failing, they may consider changing their resolver configuration to temporarily disable DNSSEC validation, or by configuring a negative trust anchor for the root zone (see RFC 7646). This should cause the problems to immediately stop.

After that, the operator should install, as soon as possible, KSK-2024 as a trust anchor and turn on DNSSEC validation again. Operators should consult their resolver software vendor for specific processes for updating trust anchors.

4. What Root Server Operators Will See

After the rollover, root server operators will start to see significantly more queries from resolvers that are unprepared for the rollover. Those queries will most likely be for the DNSKEY of the root (./IN/DNSKEY) and will also likely include queries for the DS record of the .net zone (.net/IN/DS). Additionally, since the responses can't be validated correctly, they will not be cached, which will lead to increased traffic overall from these validating resolvers. Similarly, operators of resolvers that allow other resolvers to forward through them will likely start seeing increased counts of these requests after the rollover.

Root server operators and researchers already monitor root server traffic for root DNSKEY requests to establish a baseline of how many queries per minute are typical. Root server organizations work with ICANN to share traffic statistics that allow monitoring of DNSKEY query rates. ICANN will continue to monitor those statistics after the rollover begins and will report the results to the root server operators and the rest of the DNS technical community.

Glossary

DNSSEC – Extensions to DNS that allow an authoritative server to cryptographically sign DNS records so that a resolver can be assured that the data in the records was not altered.

KSK – Key Signing Key, the key that is used to sign all the keys in a zone.

Rollover – Changing a Key Signing Key in a zone from an existing key to a newer one.

TTL – The “Time To Live” for a set of records in the DNS. When a resolver gets a set of records from an authoritative server, it usually keeps those records in its cache for the number of seconds indicated in the TTL.

Validation – Validating the signatures on the records in a zone that is protected by DNSSEC. Resolvers perform validation in order to be sure that the records they receive from an authoritative server are correct.

ZSK – Zone Signing Key, the key that is used to sign all the records in a zone other than keys (which are signed by the Key Signing Key).

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Updated on 27 July 2026 (original)
This document updates the original 17 September 2018 publication prepared for the first root KSK rollover. It has been revised for the second rollover, planned for October 2026.